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Dataset: Paper Review

1. Convert Rating to NumPy array

```
ratings = np.array(df["Rating"])
print(ratings)
```

```
[5 5 5 ... 5 5 5]
```

2. Addition

[10]
✓ 0s

```
added = ratings + 2
print("After Addition:", added)
```

```
After Addition: [7 7 7 ... 7 7 7]
```

3. Subtraction

```
subtracted = ratings - 1
print("After Subtraction:", subtracted)
```

```
After Subtraction: [4 4 4 ... 4 4 4]
```

4. Multiplication

```
multiplied = ratings * 2
print("After Multiplication:", multiplied)
```

```
After Multiplication: [10 10 10 ... 10 10 10]
```

5. Division

```
divided = ratings / 2
print("After Division:", divided)
```

```
After Division: [2.5 2.5 2.5 ... 2.5 2.5 2.5]
```

6. Mean using NumPy

```
mean_val = np.mean(ratings)
print("Mean:", mean_val)
```

```
Mean: 4.259548611111111
```

7. Median

```
median_val = np.median(ratings)
print("Median:", median_val)
```

```
Median: 5.0
```

8. Standard Deviation

```
std_val = np.std(ratings)
print("Standard Deviation:", std_val)
```

```
Standard Deviation: 1.1797609534813323
```

9. Maximum & Minimum

```python
max_val = np.max(ratings)
min_val = np.min(ratings)

print("Max:", max_val)
print("Min:", min_val)
```

```
Max: 5
Min: 1
```

10. Square of ratings

```python
squared = np.square(ratings)
print("Squared Ratings:", squared)
```

```
Squared Ratings: [25 25 25 ... 25 25 25]
```

11.Display first 5 rows

```
first_rows = df.head(5)
print("First 5 rows of dataset:")
print(first_rows)
print("Shape:", df.shape)
```

```
First 5 rows of dataset:
   Unnamed: 0                                       Product_name  \
0           0  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
1           1  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
2           2  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
3           3  DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...
4           4  DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...

                                              Review  Rating
0  Best under 60k Great performanceI got it for a...       5
1                                Good perfomence...       5
2  Great performance but usually it has also that...       5
3          My wife is so happy and best product ✌️😘       5
4  Light weight laptop with new amazing features,...       5
Shape: (2304, 4)
```

12. Display last 5 rows

```
print(df.tail())
```

```
      Unnamed: 0                                       Product_name  \
2299        2299  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2300        2300  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2301        2301  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2302        2302  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2303        2303  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...

                                              Review  Rating
2299  Great display, accurate colours at this price ...       5
2300  Superb monitor first brought 1 used for 2 mont...       5
2301                                          Awesome       5
2302                       Only one issue with adapter       5
2303  Worth the money u spend for this monitor Great...       5
```

13. Select Rating column

```python
ratings_col = df["Rating"]
print(ratings_col.head())
```

```
0    5
1    5
2    5
3    5
4    5
Name: Rating, dtype: int64
```

14. Filter ratings greater than 3

```python
filtered = df[df["Rating"] > 3]
print(filtered)
```

```
      Unnamed: 0                                       Product_name  \
0              0  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
1              1  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
2              2  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
3              3  DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...
4              4  DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...
...          ...                                                ...
2299        2299  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2300        2300  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2301        2301  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2302        2302  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...
2303        2303  MSI 27 inch Full HD IPS Panel Monitor (PRO MP2...

                                                 Review  Rating
0     Best under 60k Great performanceI got it for a...       5
1                                   Good perfomence...       5
2     Great performance but usually it has also that...       5
3              My wife is so happy and best product ✌️😘        5
4     Light weight laptop with new amazing features,...       5
...                                                 ...     ...
2299  Great display, accurate colours at this price ...       5
2300  Superb monitor first brought 1 used for 2 mont...       5
2301                                            Awesome       5
2302                        Only one issue with adapter       5
2303  Worth the money u spend for this monitor Great...       5
```

15. Count values of ratings

```python
counts = df["Rating"].value_counts()
print(counts)
```

```
Rating
5    1382
4     552
1     184
3     140
2      46
Name: count, dtype: int64
```

16. Find mean using Pandas

```python
mean_rating = df["Rating"].mean()
print("Mean Rating:", mean_rating)
```

```
Mean Rating: 4.259548611111111
```

17. Sort dataset by rating

```python
sorted_df = df.sort_values(by="Rating", ascending=False)
print(sorted_df.head())
```

```
      Unnamed: 0                                       Product_name  \
2286        2286  acer 19.5 inch HD+ LED Backlit TN Panel Monito...
2285        2285  acer 19.5 inch HD+ LED Backlit TN Panel Monito...
2283        2283  DELL P-Series 22 inch Full HD LED Backlit IPS ...
2282        2282  DELL P-Series 22 inch Full HD LED Backlit IPS ...
2281        2281  DELL P-Series 22 inch Full HD LED Backlit IPS ...

                                                 Review  Rating
2286  perfect  product with this price.  fully satis...       5
2285  Amazing work good quality display mind-blowing...       5
2283                                Good product amazing       5
2282  It's just a wow product. The quality, the buil...       5
2281  Best pick for office work like Motion graphics...       5
```

18. Add new column

```python
df["Review_Length"] = df["Review"].apply(len)
print(df.head())
```

```
   Unnamed: 0                                       Product_name  \
0           0  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
1           1  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
2           2  Lenovo Ideapad Gaming 3 Ryzen 5 Hexa Core 5600...
3           3  DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...
4           4  DELL Inspiron Athlon Dual Core 3050U - (4 GB/2...

                                              Review  Rating  Review_Length
0  Best under 60k Great performanceI got it for a...       5            324
1                                Good perfomence...       5             18
2  Great performance but usually it has also that...       5            215
3           My wife is so happy and best product ✌️😘       5             40
4  Light weight laptop with new amazing features,...       5             71
```

19. Group by rating and count

```python
grouped = df.groupby("Rating").size()
print(grouped)
```

```
Rating
1     184
2      46
3     140
4     552
5    1382
dtype: int64
```

20. Check missing values

```python
missing = df.isnull().sum()
print(missing)
```

```
Unnamed: 0       0
Product_name     0
Review           0
Rating           0
Review_Length    0
dtype: int64
```